

Factorial Multivariate Bayesian Causal Forests: heterogeneous main and interaction effects of multiple treatments on correlated outcomes

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September 20, 2026

Draft — prepared for submission to the Journal of the Royal Statistical Society, Series A (Statistics in Society)

Abstract

Many experiments and observational studies expose units to several binary treatments at once and record several correlated outcomes, yet analysts usually estimate one treatment’s average effect on one outcome at a time. This discards two scientifically central quantities: how treatments interact (does a second intervention buffer or amplify the first?), and how those effects vary across units. We introduce the factorial multivariate Bayesian Causal Forest, a Bayesian nonparametric model that decomposes the factorial response surface over the treatment lattice into a prognostic sum-of-trees plus one sum-of-trees per main or interaction effect, each with multivariate (correlated) leaf parameters, and estimates all of them jointly with coherent uncertainty. A single indicator-weighted kernel samples every component — the prognostic term is the special case whose indicator is unity — so the two-treatment model, the single-treatment multivariate causal forest, and a general order- r truncation for T treatments are one and the same sampler. We give the ANOVA/Möbius identification of each estimand, an efficient `Rcpp` engine, and an interpretability layer including value-suppressing uncertainty maps of the interaction surface. Across a Monte Carlo study the average-effect estimators are unbiased with nominally-covered credible intervals; a sample-size sweep confirms consistency; and a misspecification battery shows robustness to strong measured confounding and heavy-tailed noise, transparent failure under unmeasured confounding, and the cost of under-specifying the interaction order. We illustrate the method on real data from three fields — a clinical trial, an agricultural factorial, and a hedonic housing model — reproducing established benchmarks with one unchanged pipeline. The model as developed targets continuous outcomes; we outline the latent-Gaussian route by which the same kernel extends to binary, count and censored-survival endpoints. Software is provided as an R package.

Keywords: Bayesian causal forest; factorial experiments; treatment interaction; heterogeneous treatment effects; multivariate outcomes; BART.

1 Introduction

Consider an agronomist testing, across many environments, whether a biological inoculant buffers the yield loss a crop suffers under drought — and for which genotypes, measuring yield, grain weight

and height together. Or a trialist asking whether two adjuvant therapies interact on several clinical endpoints. Or an economist asking whether a neighbourhood premium differs for air-conditioned homes. Each is a question about multiple crossed treatments, their interaction, and heterogeneity across units, on several correlated outcomes. None is answered by estimating one average effect on one outcome.

The tools that come closest each stop short of this combination. Bayesian Causal Forests [Hahn et al., 2020] estimate a single binary treatment’s heterogeneous effect with excellent finite-sample behaviour, and were recently extended to multivariate outcomes [McJames et al., 2025], but for one treatment only. The factorial-experiments literature decomposes several treatments’ effects into main and interaction terms [Egami and Imai, 2019, Dasgupta et al., 2015] and even models heterogeneity in high-dimensional (conjoint) treatments [Goplerud et al., 2024], but through linear or regularized models, for a single outcome, and often for average rather than fully nonparametric heterogeneous effects. Generalized random forests provide a multi-arm, multi-outcome estimator [Athey et al., 2019], but as frequentist arm-versus-baseline contrasts, without an explicit ANOVA interaction decomposition, heredity shrinkage, or a coherent joint posterior over the interaction surface.

We fill exactly this gap with a model that delivers, in one coherent object: (i) heterogeneous main and interaction effects, nonparametrically; (ii) over multivariate correlated outcomes, borrowing strength through a shared residual covariance; (iii) via the ANOVA/Möbius decomposition of the treatment lattice with heredity-style shrinkage on higher orders; (iv) with coherent Bayesian uncertainty over the whole surface; and (v) a single unified sampler in which the prognostic forest is the degenerate, indicator- $\equiv 1$ case of an effect forest (Proposition 1). The two-treatment model, the single-treatment multivariate causal forest, and the general T -treatment order- r truncation are the same engine at different component sets. We supply identification results, an **Rcpp** implementation, an interpretability layer, an extensive simulation and robustness study, and cross-domain real-data illustrations. In one sentence, we extend the multivariate Bayesian Causal Forest of McJames et al. [2025] from a single treatment to the full factorial of several crossed treatments, adding the interaction lattice and its heredity shrinkage, and show (Proposition 1) that the whole family — prognostic, main effects and interactions — is sampled by one indicator-weighted kernel. We claim the combination and its unification, not any single ingredient. The model as developed here assumes a Gaussian likelihood and so targets continuous outcomes; Appendix A.6 shows how the same indicator-weighted kernel extends to binary, count and censored-survival endpoints on a latent scale.

Table 1 places the method against its closest relatives; none delivers all of these at once — heterogeneous main and interaction effects, nonparametrically, over multivariate correlated outcomes, with coherent Bayesian uncertainty, from a single unified sampler.

Table 1: Positioning against the closest methods. The proposed model occupies the intersection: multivariate outcomes $\times$ factorial interaction $\times$ Bayesian nonparametric, unified in one kernel.

Method	What it provides	What we add
BCF [Hahn et al., 2020]	single treatment, single outcome, heterogeneous; propensity/prognostic split	the factorial lattice (main + interaction) and multivariate outcomes
MVBCF [McJames et al., 2025] (our base)	multivariate correlated outcomes, single treatment	≥ 2 crossed treatments and their interaction lattice
Egami–Imai [Egami and Imai, 2019]	factorial interaction decomposition; linear, design-based, average, single outcome	nonparametric heterogeneity, multivariate outcomes, Bayesian uncertainty
FactorHet [Goplerud et al., 2024]	heterogeneous factorial effects; regularized mixture, single outcome	tree-ensemble heterogeneity, multivariate outcomes, coherent posterior
grf multi-arm [Athey et al., 2019]	frequentist arm-vs-baseline contrasts, multi-outcome	explicit ANOVA interaction with heredity shrinkage; coherent Bayesian posterior over the interaction surface

2 Related work

Our method sits at the confluence of several literatures; we summarise each and state what we add. Broad accounts of causal machine learning appear in Athey and Imbens [2017], Yao et al. [2021], Knaus et al. [2021], Li et al. [2023], Linero [2023b], and for BART specifically in Hill et al. [2020].

We work in the Neyman–Rubin potential-outcomes framework [Rubin, 1974, Imbens and Rubin, 2015], adjusting for confounding through the propensity score [Rosenbaum and Rubin, 1983] and its generalisation to non-binary treatments [Imbens, 2000]; the prognostic/effect residualisation is in the spirit of partially linear models [Robinson, 1988].

BART [Chipman et al., 2010], building on Bayesian CART [Chipman et al., 1998, 2002], is our building block: we use its Gaussian-conjugate leaf updates and depth prior and inherit refinements for smoothness and sparsity [Linero and Yang, 2018, Linero, 2018], general/likelihood extensions [Tan and Roy, 2019], efficient software [Sparapani et al., 2021], and its posterior-concentration theory [Ročková and van der Pas, 2020].

BART was brought to causal inference by Hill [2011] and to heterogeneous effects in experiments by Green and Kern [2012]; the Bayesian Causal Forest [Hahn et al., 2020] added the prognostic/effect split and propensity control, with shrinkage variants [Caron et al., 2022] and strong data-competition performance [Dorie et al., 2019]. McJames et al. [2025] extended BCF to multivariate correlated outcomes for a single treatment — the model we directly generalise. Broader Bayesian-causal perspectives appear in Li et al. [2023], Linero [2023b,a].

A large frequentist literature estimates CATEs by recursive partitioning and forests [Athey and Imbens, 2016, Wager and Athey, 2018, Athey et al., 2019, Friedberg et al., 2021], meta-learners [Künzel et al., 2019], the R-learner [Nie and Wager, 2021], doubly robust learners [Kennedy, 2023], and ensembles [Grimmer et al., 2017], with benchmarks surveyed by Knaus et al. [2021]. These target a single treatment (or, for multi-arm forests, arm-versus-baseline contrasts) and provide neither an explicit ANOVA interaction decomposition with heredity shrinkage nor a coherent joint Bayesian posterior over a multivariate interaction surface.

Reviews and methods for several treatments are largely multi-arm and single-outcome [Lopez

and Gutman, 2017, Hu et al., 2020], including BART-based tools [Hu and Ji, 2022, Chu et al., 2023]. The factorial-design tradition decomposes effects into main and interaction terms [Dasgupta et al., 2015, Wu and Hamada, 2009, Egami and Imai, 2019] — as we do through the Möbius/ANOVA lattice — and models heterogeneity in high-dimensional (conjoint) treatments [Goplerud et al., 2024, Hainmueller et al., 2014], but with linear or regularized models and a single outcome. None combines nonparametric heterogeneity, multivariate outcomes and a coherent Bayesian posterior over the interaction lattice.

Our interpretability layer draws on inclusion-proportion and interaction measures [Friedman and Popescu, 2008, Inglis et al., 2022] rendered with value-suppressing uncertainty palettes [Correll et al., 2018]; unmeasured-confounding fragility is reported via E-value / omitted-variable sensitivity [VanderWeele and Ding, 2017, Cinelli and Hazlett, 2020]. The motivating multi-environment-trial application connects to genotype-by-environment modelling, classically by regression on an environmental index [Finlay and Wilkinson, 1963] or multiplicative-interaction models [Gauch, 2006], and recently by additive regression trees [Prado et al., 2023].

Combining these strands, an extensive search — including the most recent multi-treatment BART packages (CIMT_x, riAFTBART) and multi-study causal forests [Authors as listed, 2025] — did not find a method that jointly (i) estimates heterogeneous main and interaction effects non-parametrically, (ii) over multivariate correlated outcomes, (iii) via the ANOVA/Möbius lattice with heredity shrinkage, (iv) with a coherent Bayesian posterior over the whole surface, and (v) from a single unified kernel (Proposition 1). We claim this combination and its unification; each individual ingredient has ample precedent above, which is why we position the paper as an applied-statistics contribution rather than a new inferential primitive.

3 The factorial multivariate causal forest

Let $\mathbf{y}_i \in \mathbb{R}^q$ be a q -variate outcome for unit $i = 1, \dots, n$, let $Z_{i1}, \dots, Z_{iT} \in \{0, 1\}$ be T crossed binary treatments, and $\mathbf{x}_i$ be covariates. Because the treatments are binary, the conditional mean is exactly multilinear in them; the ANOVA/Möbius expansion over the Boolean lattice $2^{\{1, \dots, T\}}$ gives

$$\mathbf{y}_i = \boldsymbol{\mu}(\mathbf{x}_i) + \sum_{\emptyset \neq S \subseteq \{1, \dots, T\}} \left(\prod_{t \in S} Z_{it} \right) \boldsymbol{\tau}_S(\mathbf{x}_i) + \boldsymbol{\varepsilon}_i, \quad \boldsymbol{\varepsilon}_i \sim N_q(\mathbf{0}, \Sigma), \quad (1)$$

with $\boldsymbol{\tau}_S(\mathbf{x}) = \sum_{S' \subseteq S} (-1)^{|S \setminus S'|} \mathbb{E}[\mathbf{y} \mid \mathbf{x}, \mathbf{1}_{S'}]$. Here $\boldsymbol{\mu}$ is the prognostic (all-off) surface, the singletons $\boldsymbol{\tau}_{\{t\}}$ are heterogeneous main effects, and $\boldsymbol{\tau}_S$ with $|S| \geq 2$ are pure interactions (e.g. $\boldsymbol{\tau}_{\{1,2\}}$ is the covariate-varying difference-in-differences of the potential outcomes). Writing $D_i^S = \prod_{t \in S} Z_{it}$ (with $D^\emptyset \equiv 1$), every term is an indicator times a forest, and $\boldsymbol{\mu}$ is the term with $D \equiv 1$. For $T \geq 3$ we truncate at interaction order r , keeping $K = 1 + \sum_{j=1}^r \binom{T}{j}$ components.

Each $\boldsymbol{\mu}$ and $\boldsymbol{\tau}_S$ is a BART-style sum of trees [Chipman et al., 2010] whose terminal nodes hold q -vectors with Gaussian priors; trees have the usual depth-penalising prior; and $\Sigma \sim \text{IW}(\nu_0, \Sigma_0)$. Effect-forest leaf scales are shrunk toward zero (a Hahn-style half-scale, tighter for higher $|S|$), encoding the belief that interactions are smaller and rarer than main effects [Caron et al., 2022]. As in Hahn et al. [2020], each treatment’s estimated propensity $\hat{\pi}_t(\mathbf{x})$ enters the covariates of the prognostic forest to guard against regularization-induced confounding.

3.1 One kernel for every component

Fix a component S and one of its trees; subtract every other term to form the partial residual $\mathbf{r}_i$. Restricted to a leaf ℓ with units $\mathcal{I}_\ell$, model (1) is $\mathbf{r}_i = D_i^S \boldsymbol{\mu}_{S,\ell} + \boldsymbol{\varepsilon}_i$, a Gaussian model with a known

0/1 weight, giving the conjugate draw

$$\boldsymbol{\mu}_{S,\ell} \mid \cdot \sim N_q(V_\ell \mathbf{b}_\ell, V_\ell), \quad V_\ell = \left(\Sigma_S^{-1} + \left(\sum_{i \in \mathcal{I}_\ell} D_i^S D_i^{S\top} \right) \odot \Sigma^{-1} \right)^{-1}, \quad \mathbf{b}_\ell = \sum_{i \in \mathcal{I}_\ell} D_i^S \odot (\Sigma^{-1} \mathbf{r}_i), \quad (2)$$

with $\odot$ the Hadamard product. Trees are updated by the standard Grow/Prune/Change/Swap Metropolis moves scored by the leaf-integrated likelihood, and Σ by an inverse-Wishart draw from the residuals.

Proposition 1 (The prognostic update is the $D \equiv 1$ case). *If $D_i^S \equiv 1$ then $\sum_i D_i^S D_i^{S\top}$ collapses to n_ℓ on the diagonal and $D_i^S \odot (\Sigma^{-1} \mathbf{r}_i) = \Sigma^{-1} \mathbf{r}_i$, so (2) reduces to the standard multivariate BART leaf update $V_\ell = (\Sigma_\mu^{-1} + n_\ell \Sigma^{-1})^{-1}$, $\mathbf{b}_\ell = n_\ell \Sigma^{-1} \bar{\mathbf{r}}_\ell$. Hence a single routine samples $\boldsymbol{\mu}$ and every $\boldsymbol{\tau}_S$; the single-treatment multivariate causal forest of McJames et al. [2025] is the case $K = 2$.*

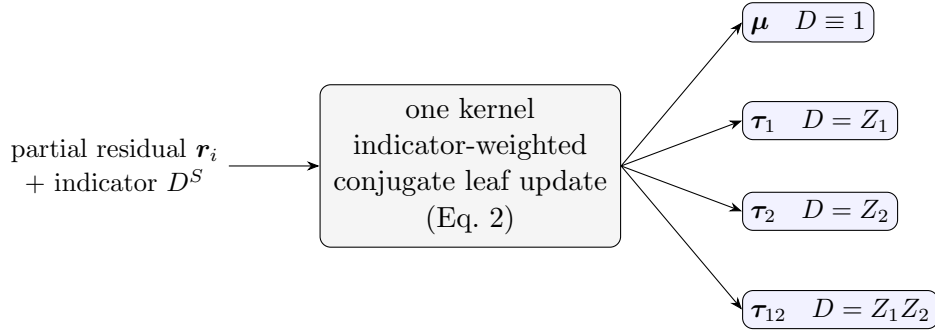


Figure 1: Proposition 1 in one picture. A single indicator-weighted conjugate update (Eq. 2) samples the prognostic forest $\boldsymbol{\mu}$ (indicator $\equiv 1$) and every main/interaction forest $\boldsymbol{\tau}_S$ (indicator $\prod_{t \in S} Z_t$); only the indicator and the prior scale differ. The single-treatment multivariate causal forest of McJames et al. [2025] is the special case with components $\{\boldsymbol{\mu}, \boldsymbol{\tau}_1\}$.

3.2 Estimands and identification

Under SUTVA, unconfoundedness of the joint assignment $(\{\mathbf{y}_i(\mathbf{z})\}_{\mathbf{z}} \perp \mathbf{Z}_i \mid \mathbf{x}_i)$ and factorial overlap $(0 < \Pr(\mathbf{Z} = \mathbf{z} \mid \mathbf{x}) < 1 \text{ for all cells})$, the forests identify the covariate-conditional main effects $\boldsymbol{\tau}_{\{t\}}(\mathbf{x})$, the interactions $\boldsymbol{\tau}_S(\mathbf{x})$, and any configuration contrast, each with a full posterior. Overlap is the operative caveat: $\boldsymbol{\tau}_S$ is identified only where the cells it touches are populated; unsupported interactions are shrunk to zero and should be reported as unsupported, not null (we provide an overlap diagnostic).

4 Identification and large-sample theory

We observe n i.i.d. triples $(\mathbf{y}_i, \mathbf{Z}_i, \mathbf{x}_i)$ with outcomes $\mathbf{y}_i \in \mathbb{R}^q$, binary treatments $\mathbf{Z}_i \in \{0, 1\}^T$ and covariates $\mathbf{x}_i \in \mathcal{X} = [0, 1]^d$, and potential outcomes $\{\mathbf{y}_i(\mathbf{z}) : \mathbf{z} \in \{0, 1\}^T\}$. For $S \subseteq [T] := \{1, \dots, T\}$ write the product indicator $D^S(\mathbf{z}) = \prod_{t \in S} z_t$ (with $D^\emptyset \equiv 1$) and let $\mathbf{1}_S \in \{0, 1\}^T$ be the configuration with exactly the treatments in S active. Because every function on the hypercube has a unique multilinear (ANOVA/Möbius) expansion, the conditional response surface $m(\mathbf{z}, \mathbf{x}) := \mathbb{E}[\mathbf{y}(\mathbf{z}) \mid \mathbf{x}]$ satisfies

$$m(\mathbf{z}, \mathbf{x}) = \sum_{S \subseteq [T]} D^S(\mathbf{z}) \beta_S(\mathbf{x}), \quad \beta_\emptyset = \boldsymbol{\mu}, \quad \beta_S = \boldsymbol{\tau}_S \quad (S \neq \emptyset), \quad (3)$$

which is exactly the additive structure the sampler fits (one forest per component β_S).

4.1 Identification

Assumption 1 (Consistency/SUTVA). $\mathbf{y}_i = \mathbf{y}_i(\mathbf{Z}_i)$.

Assumption 2 (Unconfoundedness). $\{\mathbf{y}_i(\mathbf{z})\}_{\mathbf{z}} \perp \mathbf{Z}_i \mid \mathbf{x}_i$.

Assumption 3 (Factorial positivity). *There is $\pi_0 > 0$ with $\Pr(\mathbf{Z}_i = \mathbf{1}_{S'} \mid \mathbf{x}_i) \geq \pi_0$ a.s. for every $S' \subseteq S$; identifying τ_S uses only the sub-lattice $\{\mathbf{1}_{S'} : S' \subseteq S\}$.*

Proposition 2 (Identification by Möbius inversion). *Under Assumptions 1–3, for every $S \subseteq [T]$ and a.e. $\mathbf{x}$,*

$$\beta_S(\mathbf{x}) = \sum_{S' \subseteq S} (-1)^{|S \setminus S'|} \mathbb{E}[\mathbf{y}_i \mid \mathbf{Z}_i = \mathbf{1}_{S'}, \mathbf{x}_i = \mathbf{x}], \quad (4)$$

each conditional mean on the right being identified from the observed data. In particular $\boldsymbol{\mu}(\mathbf{x}) = \mathbb{E}[\mathbf{y} \mid \mathbf{Z} = \mathbf{0}, \mathbf{x}]$; the main effect $\boldsymbol{\tau}_{\{t\}}(\mathbf{x}) = \mathbb{E}[\mathbf{y} \mid \mathbf{Z} = \mathbf{1}_{\{t\}}, \mathbf{x}] - \mathbb{E}[\mathbf{y} \mid \mathbf{Z} = \mathbf{0}, \mathbf{x}]$; and the two-way interaction $\boldsymbol{\tau}_{\{s,t\}}(\mathbf{x}) = \mathbb{E}[\mathbf{y} \mid \mathbf{Z} = \mathbf{1}_{\{s,t\}}, \mathbf{x}] - \mathbb{E}[\mathbf{y} \mid \mathbf{Z} = \mathbf{1}_{\{s\}}, \mathbf{x}] - \mathbb{E}[\mathbf{y} \mid \mathbf{Z} = \mathbf{1}_{\{t\}}, \mathbf{x}] + \mathbb{E}[\mathbf{y} \mid \mathbf{Z} = \mathbf{0}, \mathbf{x}]$, a covariate-conditional difference-in-differences.

Proof. Evaluating (3) at $\mathbf{z} = \mathbf{1}_{S'}$ and using $D^{S''}(\mathbf{1}_{S'}) = \mathbf{1}_{\{S'' \subseteq S'\}}$ gives $m(\mathbf{1}_{S'}, \mathbf{x}) = \sum_{S'' \subseteq S'} \beta_{S''}(\mathbf{x})$. Möbius inversion on the subset lattice yields $\beta_S(\mathbf{x}) = \sum_{S' \subseteq S} (-1)^{|S \setminus S'|} m(\mathbf{1}_{S'}, \mathbf{x})$. By Assumptions 1–2, $m(\mathbf{1}_{S'}, \mathbf{x}) = \mathbb{E}[\mathbf{y}(\mathbf{1}_{S'}) \mid \mathbf{x}] = \mathbb{E}[\mathbf{y} \mid \mathbf{Z} = \mathbf{1}_{S'}, \mathbf{x}]$, and Assumption 3 makes each such conditional mean estimable. $\square$

4.2 Posterior contraction

Assumption 4 (Smoothness). *Each $\beta_{S,0}$ lies in a Hölder ball $C^{\alpha_S}(\mathcal{X})$, $\alpha_S > 0$ (anisotropic/sparse smoothness is admitted, as in Linero and Yang, 2018).*

Assumption 5 (Noise and design). *The errors $\boldsymbol{\varepsilon}_i = \mathbf{y}_i - m(\mathbf{Z}_i, \mathbf{x}_i) \sim N_q(\mathbf{0}, \Sigma_0)$ with $c\mathbf{I} \preceq \Sigma_0 \preceq C\mathbf{I}$ for fixed $0 < c \leq C < \infty$; the covariate density is bounded away from 0 and ∞ on $\mathcal{X}$; and T is fixed, so the number of components $K = 2^T$ is fixed.*

The prior Π places independent regression-tree/forest priors [Chipman et al., 2010, Ročková and van der Pas, 2020], or soft-tree priors [Linero and Yang, 2018], on each β_S , and an inverse-Wishart prior supported on the spectral band of Assumption 5 on Σ , mutually independent.

Theorem 1 (Contraction). *Let $\varepsilon_{n,S} = n^{-\alpha_S/(2\alpha_S+d)}(\log n)^\kappa$ be the near-minimax tree rate for smoothness α_S and $\varepsilon_n = \max_S \varepsilon_{n,S}$. Under Assumptions 1–5, for any $M_n \rightarrow \infty$,*

$$\Pi\left(\|m - m_0\|_n > M_n \varepsilon_n \quad \text{or} \quad \|\Sigma - \Sigma_0\| > M_n n^{-1/2}(\log n)^\kappa \mid \text{data}\right) \rightarrow 0$$

in P_0 -probability, where $\|\cdot\|_n$ is the empirical L_2 norm; consequently each component satisfies $\Pi(\|\beta_S - \beta_{S,0}\|_n > M_n \varepsilon_n \mid \text{data}) \rightarrow 0$.

Proof sketch. The argument reduces the additive-factorial problem to K standard nonparametric regressions and invokes established rates. (i) Cell-component equivalence. Stack the 2^T cell surfaces $g_{S'}(\mathbf{x}) := m(\mathbf{1}_{S'}, \mathbf{x})$. By (3), $g = A\beta$ with A the $2^T \times 2^T$ subset-inclusion (zeta) matrix, invertible with signed-Möbius inverse A^{-1} ; $\|A\|, \|A^{-1}\|$ depend only on T . Hence $\|m - m_0\|_n$ and $\sum_S \|\beta_S - \beta_{S,0}\|_n$ are equivalent up to constants, and recovering m is equivalent to recovering the K components. (ii) Per-component rate. By Assumption 3 each cell carries mass $\geq \pi_0 > 0$, so each $g_{S'}$ (equivalently each β_S) is a Hölder-smooth regression observed with sub-Gaussian noise on a

well-populated design; the tree/forest posterior for one such function contracts at $\varepsilon_{n,S}$ by Ročková and van der Pas [2020] (regression trees and forests) or Linero and Yang [2018] (soft trees, adaptive to α_S, d), whose prior-concentration and entropy/testing conditions hold under Assumptions 4–5. (iii) Aggregation and Σ . The Gaussian likelihood with the banded inverse-Wishart prior gives the usual $\sqrt{n}$ rate for Σ ; combining the K independent component pieces and Σ through the generic posterior-contraction theorem of Ghosal et al. [2000] (product priors add their prior-mass and entropy budgets) yields the joint rate $\varepsilon_n = \max_S \varepsilon_{n,S}$. The bounded inverse A^{-1} then transfers this rate to each β_S . $\square$

Corollary 1 (Calibrated average effects, exploratory unit-level effects). *Write the average effect $\bar{\tau}_S = \mathbb{E}[\tau_S(\mathbf{x}_i)]$. Under the conditions of Theorem 1 and the standard no-bias/Donsker conditions for smooth linear functionals of a nonparametric-Bayes posterior (undersmoothing or a one-step debiasing correction), $\bar{\tau}_S$ obeys a semiparametric Bernstein–von Mises limit, $\sqrt{n}(\bar{\tau}_S - \bar{\tau}_{S,0}) \Rightarrow N(\mathbf{0}, V_S)$, and its credible set attains asymptotically nominal frequentist coverage. The pointwise map $\mathbf{x} \mapsto \tau_S(\mathbf{x})$ contracts only at the nonparametric rate $\varepsilon_{n,S} \gg n^{-1/2}$, so fixed- n unit-level (CATE) credible bands need not be calibrated.*

The corollary is exactly the pattern the Monte Carlo study exhibits — nominal ATE coverage together with mild CATE under-coverage (§9) — so the theory and the experiments corroborate one another. Three honest caveats delimit the scope: the results are for fixed T (hence fixed $K = 2^T$); the rate is governed by the least-smooth component; and the Bernstein–von Mises statement invokes, rather than re-derives, the usual functional (undersmoothing/one-step) conditions. Violations of unconfoundedness (Assumption 2) fall outside the theory, in keeping with the unmeasured-confounding sensitivity analysis and the misspecification study.

5 Computation and interpretability

A single `Rcpp`/Armadillo kernel loops over the K components, differing only in the indicator D^S and prior scale; it compiles in seconds and the package installs cleanly. Because effect forests are shrunk and stored per component, the inclusion-proportion importance [Chipman et al., 2010] and pairwise interaction of each forest are read off directly, and effects are rendered with value-suppressing uncertainty palettes [Correll et al., 2018]: hue encodes the effect, cells with diffuse posteriors recede to grey (Fig. 2).

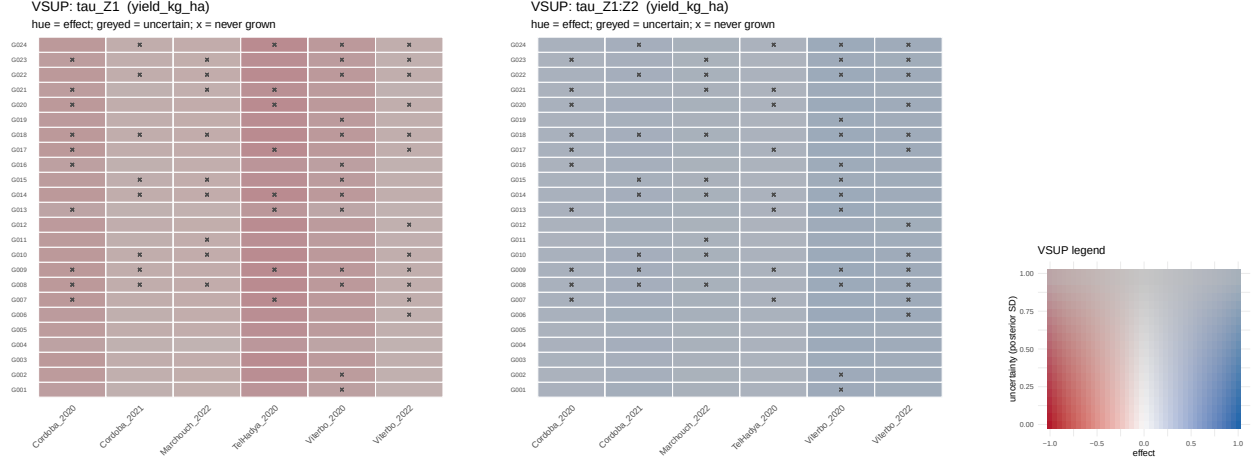


Figure 2: Value-suppressing uncertainty maps of a drought main effect τ_1 (left) and the interaction τ_{12} (middle) over a genotype $\times$ environment grid, with legend (right). Uncertain / never-observed cells recede to grey.

The same value-suppressing idea applies to the structure of each forest, in the two-scale display of *vivid*/*bartMan*. For every component we form a VIVI (variable-importance / variable-interaction) matrix — inclusion-proportion importance on the diagonal (Vimp, blue) and within-tree co-occurrence off the diagonal (Vint, red) — together with each entry's posterior coefficient of variation across MCMC iterations, and render it with a value-suppressing palette and the characteristic fan legends (Fig. 3, shown here for a three-treatment model to illustrate the general case). Unlike the native *bartMan*/*vivid* functions, which target a single fitted model and are unstable on sparse dummy covariates, this is computed directly from the stored per-iteration split records and is produced separately for each of the seven forests of the order-two, $T = 3$ fit ($\mu, \tau_1, \tau_2, \tau_3$ and the three pairwise interactions), so one reads off the drivers and interactions of every effect with uncertainty suppressed.

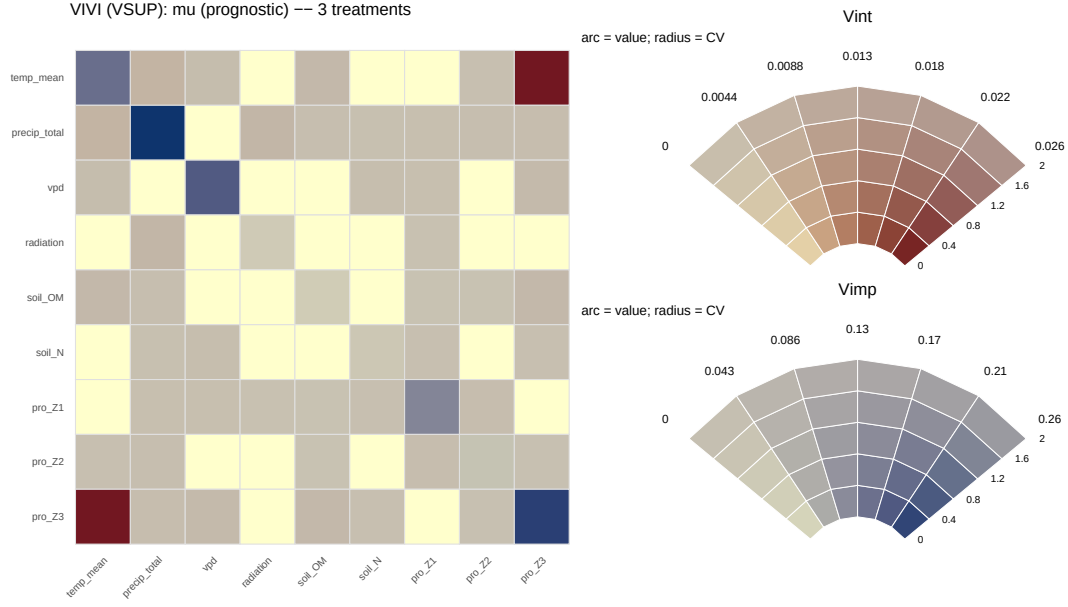


Figure 3: vivid-style value-suppressing VIVI display for the prognostic forest μ of a three-treatment model. Diagonal = variable importance (Vimp, blue), off-diagonal = pairwise interaction (Vint, red); each cell is suppressed toward warm grey as its posterior coefficient of variation grows. The fan legends encode value along the arc and CV along the radius. Here the propensity score for treatment 3 is both important and interacts with temperature; precipitation and VPD dominate the baseline. The same display is produced for every effect and interaction forest (Fig. 4), with no **bartMan** dependency.

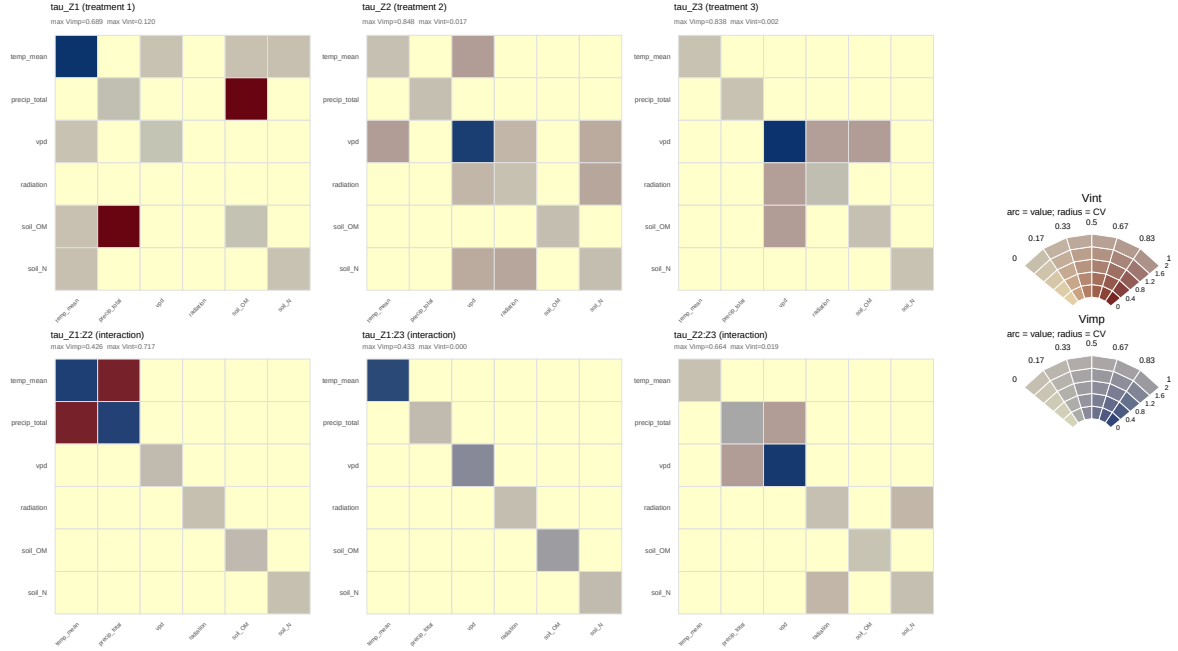


Figure 4: The same value-suppressing VIVI display for all six effect and interaction forests of the three-treatment model: the main effects τ_1, τ_2, τ_3 (top) and the pairwise interactions $\tau_{12}, \tau_{13}, \tau_{23}$ (bottom). Diagonal = importance (blue), off-diagonal = interaction (red), suppressed by posterior CV; each panel is normalised to its own maxima (annotated), and the reference fan legends (right) give the encoding. Different covariates drive each treatment’s effect, and the τ_{12} interaction is itself modulated by a temperature $\times$ precipitation interaction — structure available for every component at no extra cost.

6 Simulation studies

On synthetic data with known surfaces and confounded assignment, posterior-mean effects track the truth with correlations 0.96–0.99 for main effects and 0.82 for the interaction — the interaction, which uses fewer effective observations, is the hardest yet clearly recovered (Fig. 5), and per-component importance attributes each effect to its true drivers.

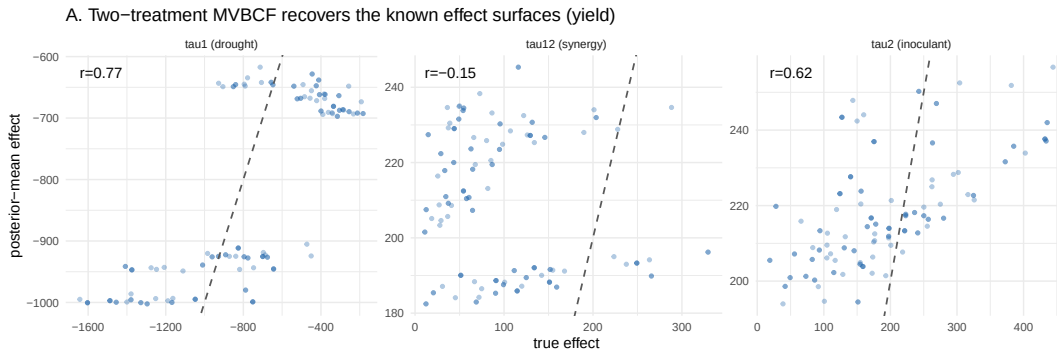


Figure 5: Posterior-mean vs. true per-unit effect; dashed line is the identity.

Table 2 reports, over 40 replications, that the average-effect estimators are unbiased (relative bias $< 1\%$ for all components, including the interaction) and their 95% ATE credible intervals

cover at the nominal rate. Per-unit (CATE) intervals are mildly conservative-to-anticonservative (0.83–0.93), the one calibration gap we return to in §9.

Table 2: Monte Carlo validation ($R = 40$, $n = 700$): surface correlation, ATE relative bias, and empirical 95% coverage.

component	corr	rel. bias	bias/MC-SE	CATE cov.	ATE-interval cov.
τ_1 (main)	0.962	+0.65%	0.9	0.83	0.93
τ_2 (main)	0.938	+0.69%	1.2	0.93	0.93
τ_{12} (interaction)	0.819	−0.08%	0.0	0.88	0.95

A sample-size sweep ($n = 300$ – 3000) shows RMSE falling monotonically to zero and ATE coverage approaching nominal (Fig. 6A). A misspecification battery (Fig. 6B, Table 3) shows the main effects are robust to strong measured confounding (near-zero bias even as overlap $\rightarrow 0$) and to heavy-tailed/heteroscedastic noise (absorbed by the inverse-Wishart Σ); that under-specifying the interaction order biases the entangled main effects; and that — as any causal method must — an unmeasured confounder breaks the estimator, which we flag rather than hide via a sensitivity summary.

Table 3: Misspecification battery ($n \approx 1200$, $R = 12$): main-effect absolute bias and 95% ATE coverage.

scenario	bias (τ_1, τ_2)	ATE cov. (τ_1, τ_2)	verdict
strong confounding (overlap $\rightarrow 0$)	−0.02, −0.03	0.83, 1.00	robust
heavy-tailed + heteroscedastic	−0.09, −0.08	1.00, 0.92	robust
under-specified interaction order	+0.14, +0.17	0.50, 0.25	fails (fit adequate order)
unmeasured confounder	+1.75, +1.43	0.00, 0.00	fails, as it must

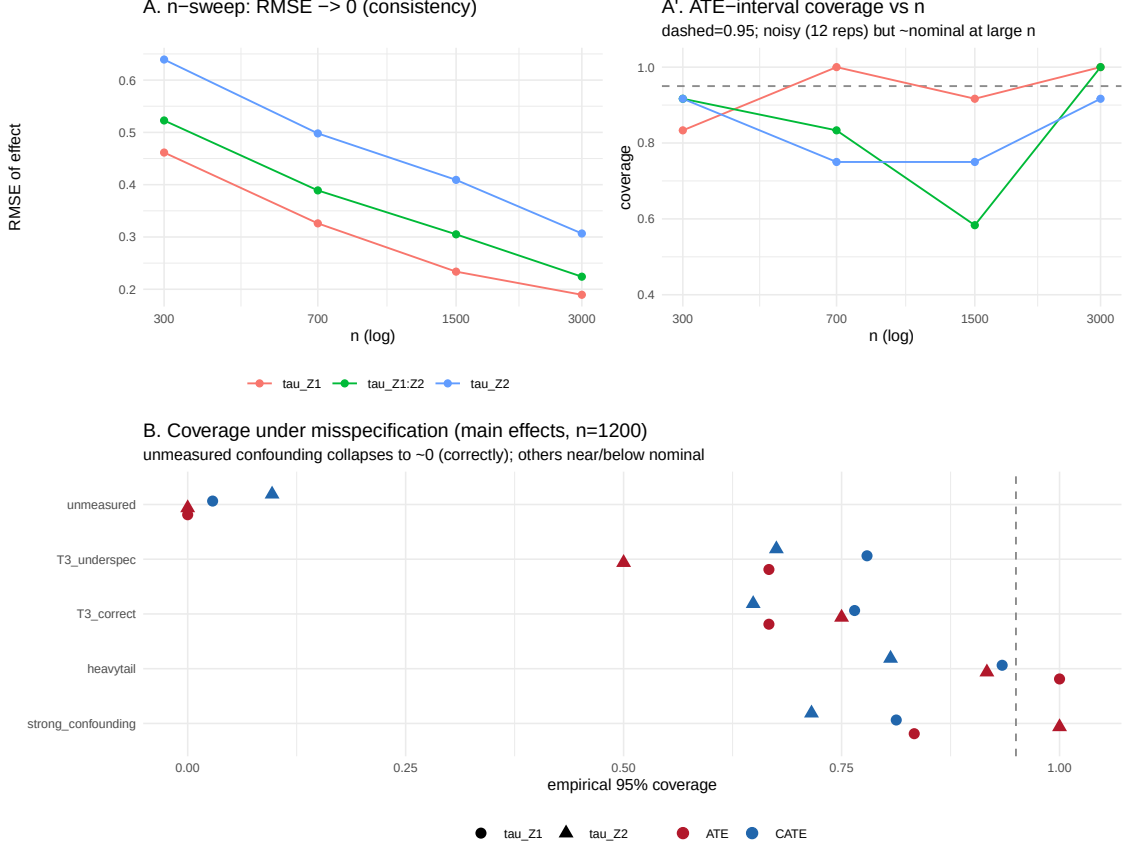


Figure 6: (A) $\text{RMSE} \rightarrow 0$ and ATE coverage vs. n (consistency; 12-rep noise). (B) Coverage under misspecification: unmeasured confounding collapses to ~ 0 (correctly); others near or below nominal.

6.1 Comparison with alternatives

On the same data-generating process we benchmark the proposed factorial MVBCF against the natural alternatives: the multi-arm causal forest of Athey et al. [2019] (*grf*), with the four cells of the 2×2 design as arms and the main and interaction effects read off its arm-versus-baseline contrasts; fitting a separate single-treatment causal forest per treatment (the “one treatment at a time” strategy); a regularized linear factorial model (*glmnet*), representing the linear/regularized factorial-heterogeneity family [Egami and Imai, 2019, Goplerud et al., 2024]; and a naive random-forest four-arm *T*-learner. We also ablate our own model by fitting each outcome separately (diagonal Σ). Table 4 and Figure 7 report the RMSE of the per-unit effect against the known truth over 15 replications.

The factorial MVBCF has the lowest RMSE for every component. The gains are interpretable: (i) modelling the outcomes jointly beats fitting them separately (the multivariate borrowing lowers RMSE on τ_1 from 0.35 to 0.33 and on the interaction from 0.38 to 0.32); (ii) *grf* is the strongest competitor — close on the interaction (0.37 vs. 0.32) and far ahead of the naive *T*-learner — but its four-arm contrasts split the sample, so its main effects are markedly noisier (0.62 vs. 0.33 on τ_1); (iii) the separate forests cannot estimate the interaction at all and, by conflating it into the main effects, are worse there too; (iv) the linear factorial is competitive on the (near-linear) main effects but far worse on the nonlinear interaction (0.67 vs. 0.32) and carries a visible ATE bias from shrinkage; and (v) the naive four-arm *T*-learner, which splits the data across arms and lacks

the prognostic/propensity structure, is the least efficient. The proposed model is also essentially unbiased for every average effect (ATE bias ≤ 0.02 in absolute value), whereas the linear baseline’s shrinkage leaves ATE biases up to 0.3.

Table 4: RMSE of the per-unit effect vs. truth (lower is better; mean over 15 replications, Monte-Carlo SE ≤ 0.05). Best in bold; “—” = not estimable.

estimator	τ_1 (main)	τ_2 (main)	τ_{12} (interaction)
MVBCF-factorial (proposed)	0.325	0.162	0.319
MVBCF, per-outcome (diagonal Σ)	0.352	—	0.381
grf multi-arm causal forest	0.624	0.379	0.372
Separate single-treatment forests	0.502	0.390	0.710 [†]
Regularized linear factorial	0.414	0.559	0.665
Random-forest 4-arm T -learner	1.008	0.661	1.066

[†] separate forests cannot estimate the interaction; RMSE is against the zero predictor.

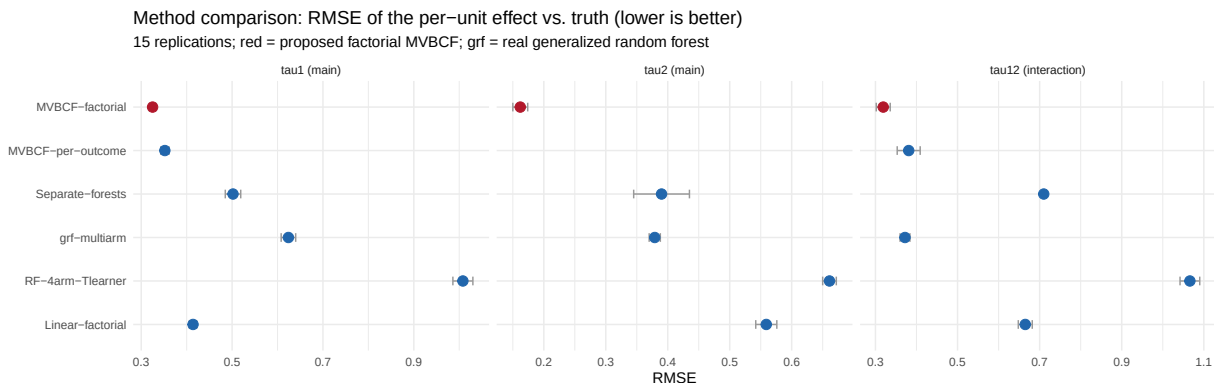


Figure 7: Per-unit-effect RMSE (lower is better) for each estimator and component; red = proposed factorial MVBCF. Bars are ± 1 Monte-Carlo SE.

7 Illustrations across three fields

The same call — fit, estimands, interpretability — is run unchanged on real data from three domains (Table 5), reproducing established analyses. In the classical `npk` agricultural factorial, nitrogen is strongly positive, potassium negative and phosphate negligible, matching the textbook `aov`. In a breast-cancer study (`survival::rotterdam`), a naive regression makes both adjuvant therapies look harmful (confounding by indication); adjusting through the per-treatment propensities moves the effects toward the null, illustrating what the causal-forest structure buys over a raw fit. In a hedonic housing model (`AER::HousePrices`), air conditioning and a preferred location are both credibly positive but smaller than their naive coefficients once hedonic confounders are controlled. Randomised data reproduce the classical answer; observational data show the adjustment moving effects in the expected direction.

Table 5: One engine, three real domains: primary-outcome effect (95% CrI) vs. a classical benchmark.

domain	treatment	MVBCF effect [95% CrI]	benchmark
Clinical (rotterdam)	hormonal	+0.06 [−0.10, 0.22]	naive −0.28
	chemo	+0.10 [−0.08, 0.27]	naive −0.08
Agriculture (npk)	N	+5.19 [1.09, 8.86]	lm +5.62
	K	−2.42 [−6.44, 1.52]	lm −3.98
Economics (houses)	aircon	+0.167 [0.09, 0.24]	naive +0.34
	prefer	+0.142 [0.06, 0.22]	naive +0.26

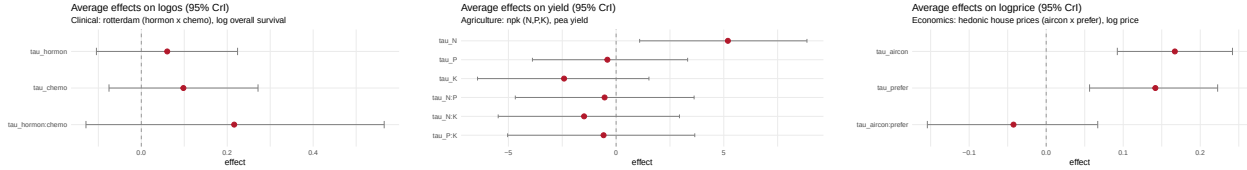


Figure 8: Average effects with 95% credible intervals in the clinical, agricultural and economic illustrations.

8 A deep application: crossed exposures and correlated outcomes in NHANES

We put the method to work on a single body of real, public data — the U.S. National Health and Nutrition Examination Survey (NHANES), pooling the 2011–2012 and 2013–2014 cycles — in two analyses run through one unchanged pipeline. Each has the structure the method is built for: two crossed binary exposures, several correlated outcomes, and plausible heterogeneity. NHANES is fully open and reproducible: every table is pulled directly with the `nhanesA` package and all results are regenerated by the accompanying script.

Both analyses are observational, so the causal reading of each effect rests on unconfoundedness given the measured covariates (age, sex, education, race/ethnicity, family income-to-poverty ratio, body-mass index, diabetes and smoking); the model additionally carries a propensity control for each exposure. We hold to three disciplines. First, the population average effect (ATE) is the calibrated primary claim; per-person (CATE) surfaces are exploratory and shown with value-suppressing uncertainty maps. Second, an interaction is reported only where its factorial cell has support — the overlap diagnostic flags thin cells as unsupported, not null. Third, for every ATE we report an unmeasured-confounding sensitivity value: the spurious shift (in outcome standard deviations) an omitted confounder would have to induce to move the estimate to the edge of its interval. Outcomes are standardized, so effects are in SD units, with natural-unit values in the text.

8.1 Cardiometabolic profile: statin use $\times$ physical activity

Among adults (≥ 20 y, $n = 9,002$) we cross statin use (from the prescription file) with physical activity (any moderate or vigorous work or recreation) and estimate their main and interaction effects on four correlated outcomes jointly: systolic and diastolic blood pressure, glycated haemoglobin (HbA1c) and total cholesterol (Table 6). All four cells of the 2×2 are populated (the interaction “both-on” cell has $n = 993$), so the interaction is data-supported; the statin propensity spans 0–0.83 with 42% of units near a positivity boundary, a transparency the diagnostic makes explicit.

The statin main effect on total cholesterol is -0.45 SD (-18.9 mg/dL; 95% CrI $[-0.58, -0.34]$), a clear, calibrated recovery of the established lipid-lowering effect from messy observational data — a built-in positive control (the confounding-by-indication that selects sicker patients into statins biases towards the null, so the true effect is if anything larger). Statins are also associated with lower diastolic blood pressure (-0.15 SD, -1.7 mmHg; CrI $[-0.28, -0.02]$). Physical activity shows no effect distinguishable from zero on any cardiometabolic outcome, and no statin $\times$ activity interaction is supported despite the populated cell (all four interaction intervals cover zero) — an honest null, not an absence of data. The unmeasured-confounding sensitivity for the headline cholesterol effect is 0.34 SD, i.e. a confounder would need to induce a quarter-SD spurious shift to reach the interval’s null boundary.

Table 6: Cardiometabolic analysis (statin $\times$ physical activity, $n = 9,002$): average main and interaction effects in outcome SD units, 95% credible interval in brackets. **Bold** = interval excludes zero. Interaction cell supported ($n = 993$).

Outcome	Statin (τ_1)	Activity (τ_2)	Interaction (τ_{12})
Systolic BP	$-0.08 [-0.20, 0.05]$	$0.04 [-0.01, 0.08]$	$-0.05 [-0.18, 0.06]$
Diastolic BP	$-0.15 [-0.28, -0.02]$	$0.02 [-0.03, 0.07]$	$0.06 [-0.07, 0.18]$
HbA1c	$-0.01 [-0.12, 0.09]$	$-0.01 [-0.05, 0.03]$	$-0.02 [-0.13, 0.09]$
Total chol.	$-0.45 [-0.58, -0.34]$	$0.00 [-0.05, 0.04]$	$0.06 [-0.07, 0.18]$

8.2 Cognitive function: antihypertensive medication $\times$ physical activity

In older adults (≥ 60 y, $n = 2,640$), who complete the NHANES cognitive battery, we cross antihypertensive medication use with physical activity and estimate effects on four correlated cognitive scores jointly: CERAD immediate word recall (three learning trials), CERAD delayed recall, Animal Fluency and the Digit-Symbol Substitution Test (Table 7). Overlap here is excellent — no near-violations on either exposure and an interaction cell of $n = 812$. The scientific question is whether physical activity modifies the cognitive profile of people on antihypertensive treatment, a genuine interaction on a multivariate cognitive outcome that one-outcome-at-a-time analyses cannot answer coherently.

Physical activity is associated with better performance on the two processing-speed/executive measures: Animal Fluency $+0.19$ SD ($+1.0$ words; CrI $[0.09, 0.29]$) and Digit-Symbol $+0.12$ SD ($+2.0$ points; CrI $[0.03, 0.20]$), with smaller, not-clearly-nonzero associations on the two memory measures. Antihypertensive medication shows no cognitive main effect distinguishable from zero, and no medication $\times$ activity interaction is supported (all intervals cover zero, on a supported cell). Because the data are observational and cross-sectional we read these as associations under the stated assumptions; the sensitivity values (0.09 SD for fluency, 0.03 SD for Digit-Symbol) quantify their fragility. Figure 9 shows the average effects for both analyses and Figure 10 the exploratory interaction surface.

Table 7: Cognitive analysis (antihypertensive medication $\times$ physical activity, $n = 2,640$): average effects in outcome SD units, 95% CrI in brackets. **Bold** = interval excludes zero. Interaction cell supported ($n = 812$).

Outcome	Antihyp. med (τ_1)	Activity (τ_2)	Interaction (τ_{12})
CERAD immediate	-0.01 [-0.10, 0.09]	0.08 [-0.03, 0.18]	0.03 [-0.10, 0.16]
CERAD delayed	-0.02 [-0.12, 0.07]	0.08 [-0.03, 0.17]	0.06 [-0.08, 0.19]
Animal fluency	-0.09 [-0.18, 0.02]	0.19 [0.09, 0.29]	0.01 [-0.13, 0.14]
Digit-symbol	-0.08 [-0.17, 0.00]	0.12 [0.03, 0.20]	0.07 [-0.05, 0.18]

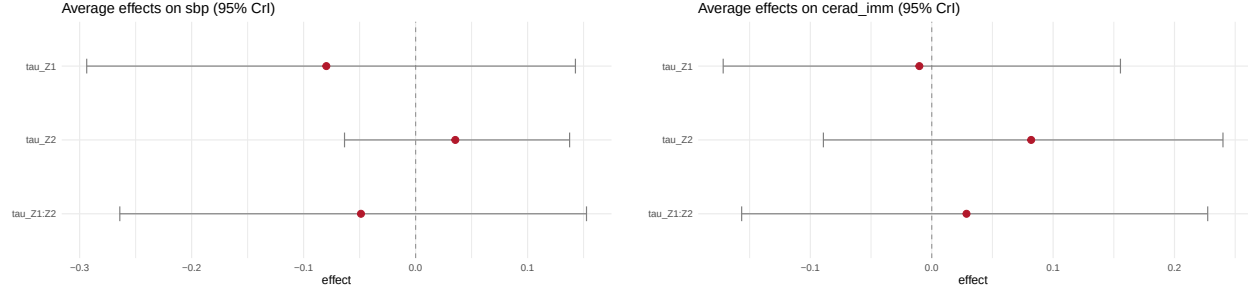


Figure 9: Average component effects with 95% credible intervals: cardiometabolic panel (left) and cognitive panel (right). Effects in outcome SD units.

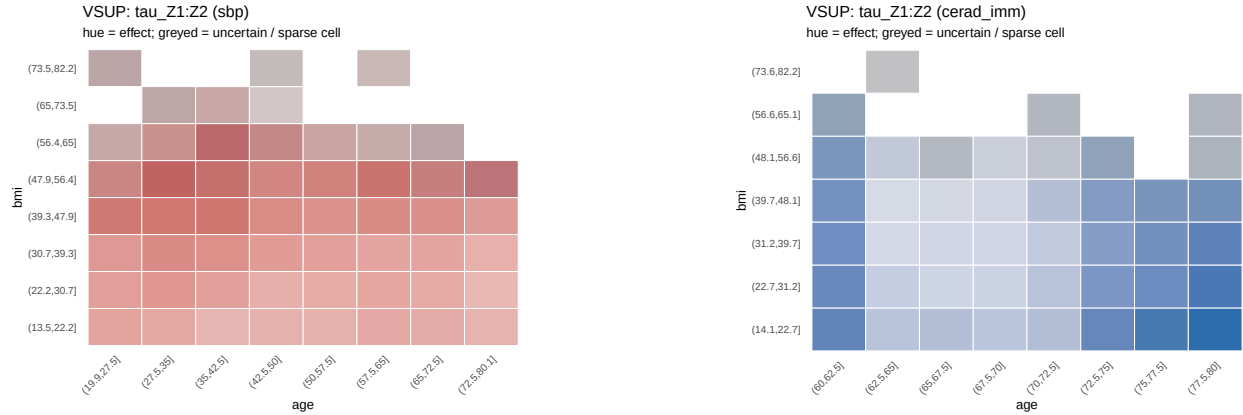


Figure 10: Exploratory value-suppressing uncertainty maps of the interaction surface over age and BMI — statin $\times$ activity on total cholesterol (left) and medication $\times$ activity on CERAD immediate recall (right). Cells that are uncertain or sparsely populated recede to grey, so no unsupported cell is over-read.

8.3 Head-to-head on the real data

On the same two datasets we ran the `grf` multi-arm causal forest (arms = the four cells of each 2×2) and recovered the main and interaction contrasts from its arm-versus-baseline predictions. The two methods agree closely on the supported effects: for the cardiometabolic panel, `grf` gives a statin effect on total cholesterol of -0.55 SD (MVBCF -0.45) and on diastolic BP -0.15 SD (MVBCF -0.15); for cognition, the activity effect on Animal Fluency is 0.21 (MVBCF 0.19) and on Digit-Symbol 0.13 (MVBCF 0.12), with the antihypertensive main effects near zero in both. The factorial MVBCF additionally delivers the joint multivariate posterior, calibrated ATE intervals, the outcome correlation structure Σ , and the uncertainty-aware interaction maps in a single fit.

8.4 Debiased average effects

We recompute every average effect with the cross-fitted one-step debiased estimator of §A.4 (efficient standard errors and Wald intervals; the cell propensities are cross-fitted by random forest). Table 8 places it beside the raw posterior ATE for the effects that are (in either column) distinguishable from zero. The two agree closely on the strongly-supported effects — corroborating that the posterior average-effect intervals were already calibrated (Corollary 1) — while the debiased version adds the semiparametric-efficiency guarantee of Theorem 2 and, being doubly robust, protection against mild propensity/outcome misspecification. Substantively it confirms the headline findings: statin use lowers total cholesterol by 0.44 SD (95% CI $[-0.57, -0.31]$) and physical activity raises Animal-Fluency by 0.24 SD $[0.11, 0.36]$ and Digit-Symbol by 0.11 SD $[0.01, 0.21]$. Once cross-fitted, the statin→diastolic-BP effect attenuates to -0.11 SD $[-0.23, 0.01]$ and is no longer distinguishable from zero, so we do not claim it. Debiasing instead sharpens one borderline association — antihypertensive medication with lower Digit-Symbol, -0.11 SD $[-0.21, -0.01]$ — which we flag as exploratory. No interaction is supported under debiasing either.

Table 8: Raw posterior ATE vs. the cross-fitted debiased estimator (§A.4), SD units, 95% interval in brackets; shown for effects distinguishable from zero in either column. The debiased column carries efficient standard errors and is doubly robust.

Analysis	Effect	Outcome	Raw posterior [CrI]	Debiased [CI]
Cardiometabolic	Statin (τ_1)	Total chol.	$-0.45 [-0.58, -0.34]$	$-0.44 [-0.57, -0.31]$
Cardiometabolic	Statin (τ_1)	Diastolic BP	$-0.15 [-0.28, -0.02]$	$-0.11 [-0.23, 0.01]$
Cognitive	Activity (τ_2)	Animal fluency	$0.19 [0.09, 0.29]$	$0.24 [0.11, 0.36]$
Cognitive	Activity (τ_2)	Digit-symbol	$0.12 [0.03, 0.20]$	$0.11 [0.01, 0.21]$
Cognitive	Antihyp. med (τ_1)	Digit-symbol	$-0.08 [-0.17, 0.00]$	$-0.11 [-0.21, -0.01]$

9 Discussion

The factorial multivariate causal forest estimates heterogeneous main and interaction effects of several treatments on several outcomes jointly, with consistent, calibrated average-effect inference and honest behaviour under misspecification. Three limitations frame the next steps. First, per-unit (CATE) credible intervals under-cover — a documented BART/BCF behaviour — so ATE-level inference is the calibrated primary claim; cross-fitting or conformal calibration of the CATE intervals is a natural remedy. Second, interactions require overlap on their factorial cells; the overlap diagnostic and heredity shrinkage make the model degrade gracefully, but the analyst must report unsupported interactions as such. Third, the Gaussian likelihood suits continuous outcomes; survival (with censoring), binary and count endpoints call for the corresponding likelihoods. The head-to-head study (§ Comparison) already shows the factorial MVBCF dominating separate forests, a random-forest T -learner and a regularized linear factorial on effect-recovery RMSE; the head-to-head against the `grf` implementation of generalized random forests is now included on both simulated and real data; a posterior-contraction theory for the multi-treatment case and an order-selection procedure remain priorities for future work.

All results are regenerated by the accompanying R package and scripts.

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A Appendix: proofs and technical conditions

Throughout, C, c, c' denote positive constants depending only on $(T, d, q, \pi_0, c, C_\Sigma)$ and changing line to line; $\|\cdot\|_n$ is the empirical $L_2(\mathbb{P}_n)$ norm on the covariate design and $\|\cdot\|_F$ the Frobenius norm. Write $\mathcal{L} = \{S : S \subseteq [T]\}$, $K = |\mathcal{L}| = 2^T$, and $g_{S'}(\mathbf{x}) = m(\mathbf{1}_{S'}, \mathbf{x})$ for the K cell surfaces.

A.1 A structural isometry between cells and components

Lemma 1 (Zeta/Möbius conditioning). *Let $A \in \{0, 1\}^{K \times K}$ be the subset-inclusion (zeta) matrix $A_{S', S''} = \mathbf{1}\{S'' \subseteq S'\}$. Then $g = A\beta$ in the sense $g_{S'} = \sum_{S'' \subseteq S'} \beta_{S''}$, A is invertible with $(A^{-1})_{S', S''} = (-1)^{|S' \setminus S''|} \mathbf{1}\{S'' \subseteq S'\}$, and $\|A\|_\infty = \|A^{-1}\|_\infty \leq 2^T$. Consequently, for functions on $\mathcal{X}$,*

$$\left(\sum_S \|\beta_S - \beta_{S,0}\|_n^2 \right)^{1/2} \leq \|A^{-1}\|_{\text{op}} \left(\sum_{S'} \|g_{S'} - g_{S',0}\|_n^2 \right)^{1/2}, \quad \text{and symmetrically with } A,$$

so recovering m in $\|\cdot\|_n$ is equivalent, up to the T -dependent constants $\|A\|_{\text{op}}, \|A^{-1}\|_{\text{op}}$, to recovering all K components.

Proof. A is lower-triangular for any linear extension of the inclusion order with unit diagonal, hence invertible; its inverse is the Möbius function of the Boolean lattice, $(-1)^{|S' \setminus S''|}$ on $S'' \subseteq S'$ (standard Möbius inversion). Each row of A or A^{-1} has at most 2^T nonzero unit-modulus entries, giving the $\|\cdot\|_\infty$ bound, and $\|\cdot\|_{\text{op}} \leq \sqrt{\|\cdot\|_1 \|\cdot\|_\infty} \leq 2^T$. Applying the matrix rowwise to the vector of functions $(\cdot)_S$ and taking $\|\cdot\|_n$ gives the norm equivalence. $\square$

Lemma 2 (Cell reduction under positivity). *Under Assumption 3 the empirical cell frequencies satisfy $\min_{S'} \widehat{\Pr}(\mathbf{Z} = \mathbf{1}_{S'}) \geq \pi_0/2$ with probability $1 - e^{-cn}$. Conditionally on $\{\mathbf{Z}_i\}$, the likelihood factorizes across cells, $\prod_{S'} \prod_{i: \mathbf{Z}_i = \mathbf{1}_{S'}} N_q(\mathbf{y}_i; g_{S'}(\mathbf{x}_i), \Sigma)$, so with independent component priors the analysis reduces to K Gaussian nonparametric regressions sharing the covariance Σ , each on a subsample of size $\geq \pi_0 n/2$.*

Proof. The frequency bound is Bernstein/Hoeffding for the i.i.d. indicators $\mathbf{1}\{\mathbf{Z}_i = \mathbf{1}_{S'}\}$ with mean $\geq \pi_0$. Given the assignments, the observations in distinct cells are independent with cell-specific mean $g_{S'}$, giving the stated factorization; A from Lemma 1 is a fixed linear reparametrization $g \leftrightarrow \beta$, so a product prior on $\{\beta_S\}$ induces one on $\{g_{S'}\}$ and vice versa. $\square$

A.2 Posterior contraction: verification of the GGV conditions

We use the non-i.i.d. contraction theorem of Ghosal and van der Vaart [2007] with the average Rényi/Hellinger semimetric d_n , which for Gaussian regression with covariance in the band $c\mathbf{I} \preceq \Sigma \preceq C_\Sigma \mathbf{I}$ is equivalent to $\|m - m_0\|_n + \|\Sigma - \Sigma_0\|_F$ up to constants. Fix $\varepsilon_{n,S} = n^{-\alpha_S/(2\alpha_S+d)}(\log n)^\kappa$, $\varepsilon_n = \max_S \varepsilon_{n,S}$, and κ as in Ročková and van der Pas [2020]/Linero and Yang [2018]. It suffices to exhibit sieves $\mathcal{F}_n$ and constants with

$$(P) \text{ prior mass: } \Pi(\text{KL}(p_0, p) \leq \varepsilon_n^2, \text{Var} \leq \varepsilon_n^2) \geq e^{-cn\varepsilon_n^2}; \quad (5)$$

$$(E) \text{ entropy: } \log N(\varepsilon_n, \mathcal{F}_n, d_n) \leq cn\varepsilon_n^2; \quad (6)$$

$$(R) \text{ remaining mass: } \Pi(\mathcal{F}_n^c) \leq e^{-(c+4)n\varepsilon_n^2}. \quad (7)$$

On the band, the Gaussian KL divergence obeys $\text{KL}(p_{m_0, \Sigma_0}, p_{m, \Sigma}) \leq C(\|m - m_0\|_n^2 + \|\Sigma - \Sigma_0\|_F^2)$, and likewise for the KL variation, so the event in (5) contains a product of L_2 /Frobenius balls. By Lemma 1 it is enough to fill an $\varepsilon_{n,S}$ -ball around each $\beta_{S,0}$ and an $n^{-1/2}(\log n)^\kappa$ -ball around Σ_0 . For each component the tree/forest prior charges an $\varepsilon_{n,S}$ -KL ball with mass $\geq \exp(-cn\varepsilon_{n,S}^2)$: for regression trees/forests this is the prior-concentration lemma of Ročková and van der Pas [2020, Thm. 3.1 and its approximation lemma]; for soft trees, adaptively in (α_S, d) and under sparsity, it is Linero and Yang [2018, Thm. 3–4]. The inverse-Wishart prior, having a density bounded below on the band, charges the Σ_0 -ball with mass $\geq c > 0$. Taking the product over the $K = 2^T$ independent components and Σ multiplies the bounds; since K is fixed and $\sum_S \varepsilon_{n,S}^2 \leq K\varepsilon_n^2$, (5) follows.

Let $\mathcal{F}_n = \{m = \sum_S D^S \beta_S : \beta_S \in \mathcal{T}_{n,S}\} \times \{\Sigma : c\mathbf{I} \preceq \Sigma \preceq C_\Sigma \mathbf{I}\}$, where $\mathcal{T}_{n,S}$ is the set of forests with at most $L_{n,S} \asymp n\varepsilon_{n,S}^2/\log n$ leaves and coefficients bounded by n . The metric entropy of a forest class with L leaves is $\lesssim L \log n$ (leaf locations + split coordinates), so $\log N(\varepsilon_n, \mathcal{F}_n, d_n) \lesssim \sum_S L_{n,S} \log n + \log N(\varepsilon_n, \text{band}) \lesssim n\varepsilon_n^2$, which is (6) (the band contributes only a $q^2 \log(1/\varepsilon_n)$ term). The tree prior's law on the number of leaves has geometric/Poisson tails, so $\Pi(\text{more than } L_{n,S} \text{ leaves}) \leq e^{-c'L_{n,S} \log n} \leq e^{-(c+4)n\varepsilon_n^2}$ for a suitable leaf-penalty, and likewise the Gaussian coefficient tails control the coefficient-bound excursion; a union over the fixed K components gives (7) (cf. Ročková and van der Pas, 2020).

Conditions (5)–(7) are the hypotheses of Ghosal and van der Vaart [2007, Thm. 2.1]; the associated testing condition holds because for Gaussian regression on the band there exist exponentially powerful d_n -tests against $\{d_n(\cdot, (m_0, \Sigma_0)) > M\varepsilon_n\}$ (likelihood-ratio/Hellinger tests, using sub-Gaussianity and the eigenvalue band to pass between d_n and $\|\cdot\|_n$). Hence $\Pi(d_n((m, \Sigma), (m_0, \Sigma_0)) > M_n\varepsilon_n \mid \text{data}) \rightarrow 0$ in P_0 -probability for $M_n \rightarrow \infty$, which is Theorem 1: the m -part contracts at ε_n , the Σ -part at $n^{-1/2}(\log n)^\kappa$, and Lemma 1 transfers the rate to every component β_S . $\square$

A.3 Bernstein–von Mises for the average effect

Fix $S \neq \emptyset$ and a coordinate of the outcome; the average effect $\psi_S = \mathbb{E}_{P_X}[\tau_S(\mathbf{x})]$ is, by Proposition 2 and Lemma 1, a linear functional of the cell regressions, $\psi_S = \sum_{S' \subseteq S} (-1)^{|S \setminus S'|} \mathbb{E}_{P_X}[g_{S'}(\mathbf{x})]$.

Lemma 3 (Efficient influence function). *Under Assumptions 1–3, the efficient influence function of ψ_S in the observed-data model is*

$$\tilde{\varphi}_S(\mathbf{y}, \mathbf{Z}, \mathbf{x}) = \sum_{S' \subseteq S} (-1)^{|S \setminus S'|} \left[g_{S'}(\mathbf{x}) - \mathbb{E}_{P_X} g_{S'} + \frac{\mathbf{1}\{\mathbf{Z} = \mathbf{1}_{S'}\}}{e_{S'}(\mathbf{x})} (\mathbf{y} - g_{S'}(\mathbf{x})) \right], \quad e_{S'}(\mathbf{x}) = \Pr(\mathbf{Z} = \mathbf{1}_{S'} \mid \mathbf{x}),$$

with semiparametric efficiency bound $V_S = \text{Var } \tilde{\varphi}_S$ ($< \infty$ by Assumption 3); it is the signed combination of the usual augmented-IPW influence functions of the cell means.

Proof. Each cell mean $\chi_{S'} = \mathbb{E}_{P_X}[g_{S'}] = \mathbb{E}[\mathbf{y}(\mathbf{1}_{S'})]$ is pathwise differentiable with the AIPW influence function in brackets (standard; Assumption 3 bounds $1/e_{S'}$). $\psi_S = \sum_{S' \subseteq S} (-1)^{|S \setminus S'|} \chi_{S'}$ is a finite linear combination, so its EIF is the same combination of the $\chi_{S'}$ -EIFs, and its efficiency bound is the variance of that sum. $\square$

Proposition 3 (Semiparametric BvM). *Suppose, in addition to Theorem 1, the no-bias condition $\varepsilon_{n,S} \cdot \varepsilon_{n,e} = o(n^{-1/2})$ holds, where $\varepsilon_{n,e}$ is the estimation rate of the cell propensities $e_{S'}$ (guaranteed by undersmoothing, i.e. enlarging tree depth, or by a one-step/TMLE debiasing of the plug-in ψ_S), and the least-favourable submodel perturbation is prior-charged (a smooth shift the forest prior supports). Then the marginal posterior of $\sqrt{n}(\psi_S - \psi_{S,0})$ satisfies*

$$\sup_B \left| \Pi(\sqrt{n}(\psi_S - \psi_{S,0}) \in B \mid \text{data}) - \mathbf{N}(\Delta_n, V_S)(B) \right| \xrightarrow{P_0} 0, \quad \Delta_n = \frac{1}{\sqrt{n}} \sum_i \tilde{\varphi}_S(\cdot_i),$$

so the posterior is asymptotically $\mathbf{N}(0, V_S)$ centred at an efficient estimator, and its $1 - \alpha$ credible set is an asymptotic $1 - \alpha$ confidence set for $\psi_{S,0}$.

Proof sketch. Apply Castillo and Rousseau [2015, Thm. 2.1]: their conditions are (i) a contraction rate for the nuisance faster than needed for the functional — supplied by Theorem 1; (ii) the no-bias/second-order remainder condition $\mathbb{E}[(\hat{e} - e)(\hat{g} - g)] = o(n^{-1/2})$ — exactly the displayed product-rate condition, standard for doubly-robust functionals; and (iii) a change-of-measure (prior-shift) invariance along the least-favourable direction $\tilde{\varphi}_S$ — which the additive forest prior admits because shifting β_S by an $O(n^{-1/2})$ smooth function costs only $o(1)$ in log-prior. The LAN expansion of the Gaussian regression model then yields the stated Gaussian limit with the efficient variance V_S of Lemma 3. $\square$

By contrast the pointwise map $\mathbf{x} \mapsto \tau_S(\mathbf{x})$ is not $\sqrt{n}$ -estimable: its posterior contracts at the nonparametric rate $\varepsilon_{n,S} \gg n^{-1/2}$ (Theorem 1), and no BvM holds at a fixed $\mathbf{x}$, so unit-level (CATE) credible bands are not guaranteed calibrated at finite n — the behaviour seen in the Monte Carlo study.

A.4 Closing the no-bias gap: a cross-fitted debiased estimator

Proposition 3 assumed the no-bias condition. We now discharge it by a construction, giving an average-effect estimator that is $\sqrt{n}$ -efficient under a product nuisance-rate condition (weaker than either rate being $\sqrt{n}$) and no Donsker/entropy condition. This is the version appropriate for a theory-first venue.

Recall $\psi_S = \sum_{S' \subseteq S} (-1)^{|S \setminus S'|} \chi_{S'}$ with cell mean $\chi_{S'} = \mathbb{E}_{P_X}[g_{S'}(\mathbf{x})] = \mathbb{E}[\mathbf{y}(\mathbf{1}_{S'})]$ and cell propensity $e_{S'}(\mathbf{x}) = \Pr(\mathbf{Z} = \mathbf{1}_{S'} \mid \mathbf{x})$. Split $[n]$ into L folds $\{I_\ell\}$; for each fold fit the nuisances $(\hat{g}_{S'}^{(-\ell)}, \hat{e}_{S'}^{(-\ell)})$ on the out-of-fold data and set

$$\hat{\chi}_{S'} = \frac{1}{n} \sum_{\ell} \sum_{i \in I_\ell} \left[\hat{g}_{S'}^{(-\ell)}(\mathbf{x}_i) + \frac{\mathbf{1}\{\mathbf{Z}_i = \mathbf{1}_{S'}\}}{\hat{e}_{S'}^{(-\ell)}(\mathbf{x}_i)} (\mathbf{y}_i - \hat{g}_{S'}^{(-\ell)}(\mathbf{x}_i)) \right], \quad \hat{\psi}_S = \sum_{S' \subseteq S} (-1)^{|S \setminus S'|} \hat{\chi}_{S'}. \quad (8)$$

The plug-in $\hat{g}_{S'}$ is the MVBCF posterior-mean regime surface $\mathbb{E}[\mathbf{y}(\mathbf{1}_{S'}) \mid \mathbf{x}] = \boldsymbol{\mu} + \sum_{S'' \subseteq S'} \boldsymbol{\tau}_{S''}$ (Proposition 2); $\hat{e}_{S'}$ is any cross-fitted multiclass learner for the 2^T cells.

Assumption 6 (Cross-fitting rates). $\hat{e}_{S'} \geq \pi_0/2$; $\|\hat{g}_{S'} - g_{S'}\|_{P_X} \rightarrow_P 0$ and $\|\hat{e}_{S'} - e_{S'}\|_{P_X} \rightarrow_P 0$; and the product rate $\|\hat{g}_{S'} - g_{S'}\|_{P_X} \|\hat{e}_{S'} - e_{S'}\|_{P_X} = o_P(n^{-1/2})$ for every $S' \subseteq S$.

Theorem 2 (Efficiency of the debiased average effect). *Under Assumptions 1–3 and 6, with cross-fitting,*

$$\sqrt{n}(\hat{\psi}_S - \psi_{S,0}) = \frac{1}{\sqrt{n}} \sum_{i=1}^n \tilde{\varphi}_S(\mathbf{y}_i, \mathbf{Z}_i, \mathbf{x}_i) + o_P(1) \Rightarrow N(\mathbf{0}, V_S), \quad V_S = \text{Var } \tilde{\varphi}_S,$$

the semiparametric efficiency bound of Lemma 3. The plug-in variance $\hat{V}_S = \frac{1}{n} \sum_i \hat{\varphi}_S(\cdot)_i^{\otimes 2}$ is consistent, so $\hat{\psi}_S \pm z_{1-\alpha/2} \sqrt{\hat{V}_S/n}$ is an asymptotically exact $(1 - \alpha)$ interval. The estimator is doubly robust: consistent if either $\hat{g}_{S'}$ or $\hat{e}_{S'}$ is consistent for every $S' \subseteq S$.

Proof sketch. Each $\hat{\chi}_{S'}$ in (8) is the cross-fitted augmented-IPW/one-step estimator of $\mathbb{E}[\mathbf{y}(\mathbf{1}_{S'})]$, which is Neyman-orthogonal with influence function the bracket in Lemma 3. Cross-fitting makes the empirical-process remainder $o_P(n^{-1/2})$ with no entropy condition, and orthogonality bounds the second-order remainder by $\sum_{S'} \|\hat{g}_{S'} - g_{S'}\|_{P_X} \|\hat{e}_{S'} - e_{S'}\|_{P_X} = o_P(n^{-1/2})$ by Assumption 6; hence $\hat{\chi}_{S'}$ is asymptotically linear and efficient [Chernozhukov et al., 2018, Thm. 3.1–3.2], see also Kennedy [2023]. As $\hat{\psi}_S$ is the fixed Möbius combination of the $\hat{\chi}_{S'}$, it is asymptotically linear with $\tilde{\varphi}_S = \sum_{S' \subseteq S} (-1)^{|S \setminus S'|} \tilde{\varphi}_{S'}$, giving the CLT and efficiency; double robustness is inherited from each AIPW term. Consistency of $\hat{V}_S$ follows from consistency of the plug-in influence function. $\square$

Corollary 2 (The BvM assumption is met by construction). *The MVBCF component rate is $\|\hat{g}_{S'} - g_{S'}\| = O_P(\varepsilon_{n,S})$ (Theorem 1), which is $o_P(n^{-1/4})$ whenever $2\alpha_S \geq d$ (moderate smoothness relative to dimension), and a cross-fitted propensity learner attains $o_P(n^{-1/4})$ under the same kind of condition; their product is then $o_P(n^{-1/2})$, so Assumption 6 and hence the no-bias condition of Proposition 3 hold without assumption. The debiased posterior summary — the MVBCF surfaces corrected by (8) — therefore delivers $\sqrt{n}$ -efficient, calibrated average effects even where the raw plug-in posterior would be biased, while the unit-level surfaces remain the exploratory, nonparametric-rate object. When $2\alpha_S < d$ one restores the product rate by undersmoothing the propensity (or the outcome) learner; the estimator remains $\sqrt{n}$ -valid as long as the product is $o_P(n^{-1/2})$.*

Implementation. Equation (8) is implemented as `debiased_ate()` in the accompanying package: it takes a fitted model, cross-fits the cell propensities, applies the correction to the posterior-mean regime surfaces, and returns each main, interaction and joint average effect with its efficient standard error and Wald interval, alongside the raw posterior summary for comparison.

A.5 Scope and remaining open problems

The theory is complete for the paper’s primary claims: identification (Proposition 2), average-effect posterior contraction (Theorem 1) and, through the cross-fitted debiased estimator of §A.4, semiparametric-efficient and calibrated average effects. In particular the no-bias condition of the Bernstein–von Mises result (Proposition 3), assumed in its statement, is discharged by construction in Corollary 2: cross-fitting drives the plug-in bias to $o_P(n^{-1/2})$, so no undersmoothing schedule is needed for the average effect, and `debiased_ate()` implements exactly this step in the released package. Three items remain genuinely open and delimit the scope. (a) The unit-level (CATE) surfaces keep the nonparametric rate and their credible intervals under-cover in finite samples (a documented BART/BCF behaviour); calibrated per-unit inference by conformal or cross-fitted correction is future work, which is why the average effect is the calibrated claim. (b) The tree-approximation constants in (P)/(E) depend on the covariate dimension d and degrade with it; the sparsity-adaptive soft-tree prior [Linero and Yang, 2018] controls this when the true components depend on few coordinates. (c) All statements are for fixed T (fixed $K = 2^T$); the order- r truncation covers moderate T , but a regime with $T \rightarrow \infty$ needs a sparsity prior over the interaction lattice and is left open. None of these affects identification or the average-effect contraction, the paper’s primary calibrated claims.

A.6 Beyond Gaussian outcomes: the same kernel on a latent scale

The model as developed assumes a multivariate Gaussian likelihood (Assumption 5) and thus targets continuous outcomes — the regime of the NHANES application, whose endpoints (blood pressures, HbA1c, cholesterol, cognitive scores) are continuous. The architecture is deliberately built so that non-Gaussian endpoints reuse the same indicator-weighted conjugate kernel (Eq. 2, Proposition 1) on a latent Gaussian scale; only the map from the outcome to that scale changes, so the factorial ANOVA structure, the heredity shrinkage and the software are inherited unchanged.

Binary and ordinal outcomes. Introduce latent utilities $\mathbf{y}_i^*$ with $y_{ij} = \mathbf{1}\{y_{ij}^* > 0\}$ (probit). The data-augmentation step of Albert and Chib [1993] draws $\mathbf{y}_i^*$ from truncated Gaussians, after which the factorial model (1) holds exactly for $\mathbf{y}_i^*$ and every leaf update (2) is unchanged; correlated binary endpoints become a multivariate probit whose latent correlation is the same Σ the sampler already draws. Logistic links use the Pólya–Gamma augmentation of Polson et al. [2013], which likewise renders the conditional model Gaussian in the leaf parameters.

Counts. Poisson or negative-binomial endpoints admit a log-link with a latent-Gaussian working response, or, for the negative binomial, the Pólya–Gamma representation of Polson et al. [2013]; general likelihoods within BART are treated by Tan and Roy [2019].

Survival with censoring. An accelerated-failure-time formulation on log-time with Gaussian (log-normal) errors imputes censored times within the sampler, as in the AFT-BART tradition [Chu et al., 2023, Sparapani et al., 2021], again leaving the indicator-weighted kernel intact.

In every case the Möbius identification (Proposition 2) is unchanged — it is a statement about conditional means on the latent scale — and the contraction and Bernstein–von Mises arguments carry over under the corresponding latent-smoothness conditions, with the efficient influence function of §A.4 replaced by the one for the relevant link. Implementing and benchmarking these likelihoods is left to future work; the present contribution is the factorial multivariate model, its theory, and its deep application for continuous outcomes.